ARAB YOUTH SPACE HACKATHON 2026 · 813 CHALLENGE · PROOF OF CONCEPT

Doha Flood Watch

Mapping where heavy rain hit Doha, using free Sentinel-1 radar images

TEAM	COUNTRY	THEME
Stark	Qatar	Climate-induced Disasters

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THE STORY

What happened after the 2018 storm

1

20 Oct 2018

Doha gets roughly a year's rain in one day. Roads, underpasses and buildings flood.

2

22 Oct

Optical satellite passes over, but scattered clouds and shadows hide the ground.

3

27 Oct

Sentinel-1 radar passes over: it sees through cloud, day or night.

4

~1 minute later

Our notebook turns the images into a map of the city and a ranked list of neighbourhoods.

Radar sees through the cloud that blocked the optical satellite, and the data is free.

THE PROBLEM

Why heavy rain floods Doha

~75 mm

That's Qatar's average rain for a whole year, so drainage was built for light rain.

1 day

On 20 Oct 2018 Doha received roughly a year's rainfall. Roads, underpasses and buildings flooded.

Flat

The city is paved and the desert ground is hard, so water collects in streets and low ground.

It affects residents, commuters, schools and businesses, and the agencies that have to find flooded areas and pump them out.

BUSINESS USE CASE

Which areas get crews first?

Ashghal (Public Works Authority)

Decides: where to send pumps; which drainage to upgrade.
Today: hotline calls and field patrols.

Ministry of Municipality

Decides: where new development sits on flood-prone ground.
Today: historic complaints, no citywide map.

Civil Defence / Ministry of Interior

Decides: where to respond after a storm.
Today: reports and social media, slow to assemble.

Insurers in Qatar

Decides: which claims to triage; flood risk per location.
Today: claim-by-claim inspection.

We give them a map of the whole city after each satellite pass, even through cloud, as an image, a web map and tables they can open in GIS or Excel.

DATA

Data we used

Dataset	Dates	Role
Sentinel-1 radar (RTC, VV)	3, 15 and 27 Oct 2018	Dry baseline, before and after the storm
Sentinel-2 optical (L2A)	12 and 22 Oct 2018	True-colour map; proof that optical fails
ESA WorldCover 10 m	2021 map	Land cover for exposure, sea mask

Area of interest

Central Doha, Al Rayyan and northern Al Wakrah: about 22 × 22 km at 20 m resolution.

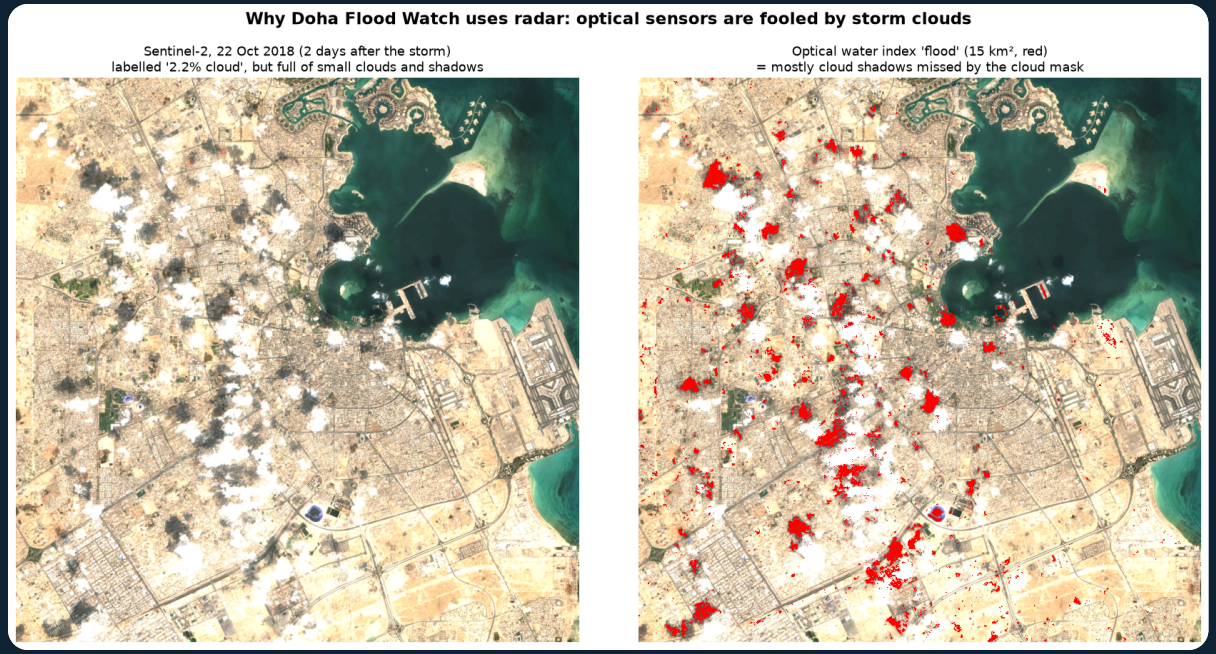
Hyperspectral

We had no clear scene after the storm, so it isn't used here. We plan to add 813 or EnMAP data in incubation.

All via Microsoft Planetary Computer, no account needed · Copernicus Sentinel data · WorldCover CC BY 4.0

WHY RADAR

Why we switched to radar



The "2.2% cloud" image two days after the storm is full of small clouds over Doha.

An optical water index flags **15 km² of "flood"**, mostly cloud shadows.

Sentinel-1 radar sees through cloud, day and night.

Reproduce: python src/make_why_radar_figure.py

APPROACH

How it works

1 · Load

Sentinel-1 before, after and dry-week images, clipped to Doha at 20 m

2 · Clean

Median speckle filter; remove sea plus a 300 m shoreline buffer

3 · Detect

Backscatter up > 3 dB = saturated ground; down + dark = standing water

4 · Validate

Same detector on a dry week gives the false-alarm rate; thresholds 2 to 5 dB

5 · Rank

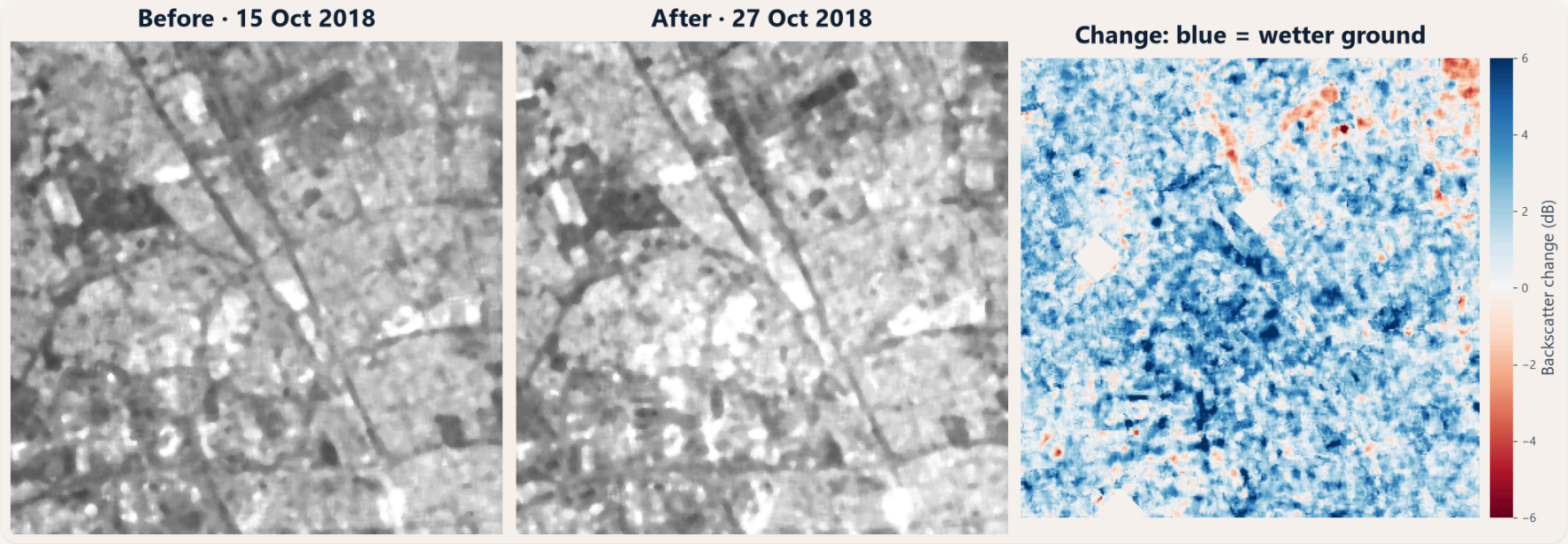
km² by land cover; 1 km cells ranked by % of land saturated

The notebook saves a map image, an interactive web map and CSV tables. It runs in about a minute on a laptop, with no API keys.

Code: src/floodmap.py · notebooks/01_doha_flood_mapping.ipynb

HOW RADAR SEES RAIN

What the radar shows

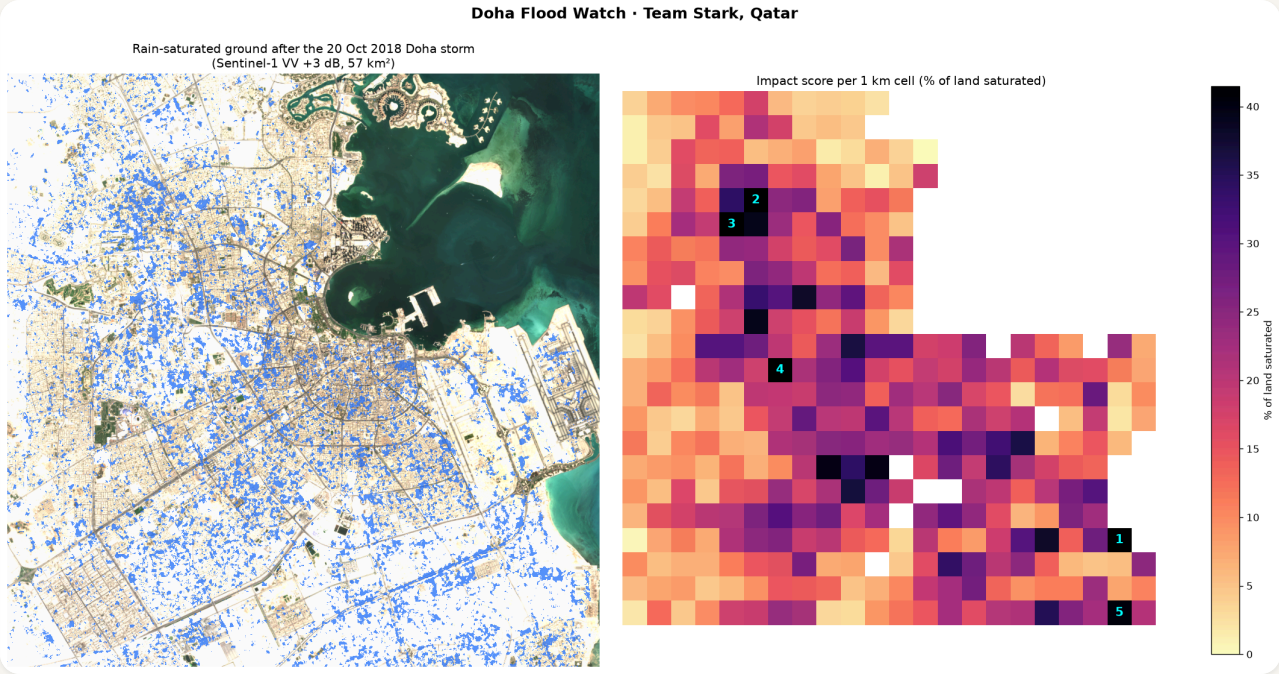


6 × 6 km of western Doha. Water raises the soil's dielectric constant, so saturated ground reflects more radar energy back to the satellite: **blue = wetter after the storm**.

Sentinel-1A, same orbit and time of day for both images

RESULT

Where the 20 Oct 2018 storm hit Doha



Blue: ground that became saturated (radar +3 dB)

Grid: % of each 1 km cell's land saturated; darker = worse

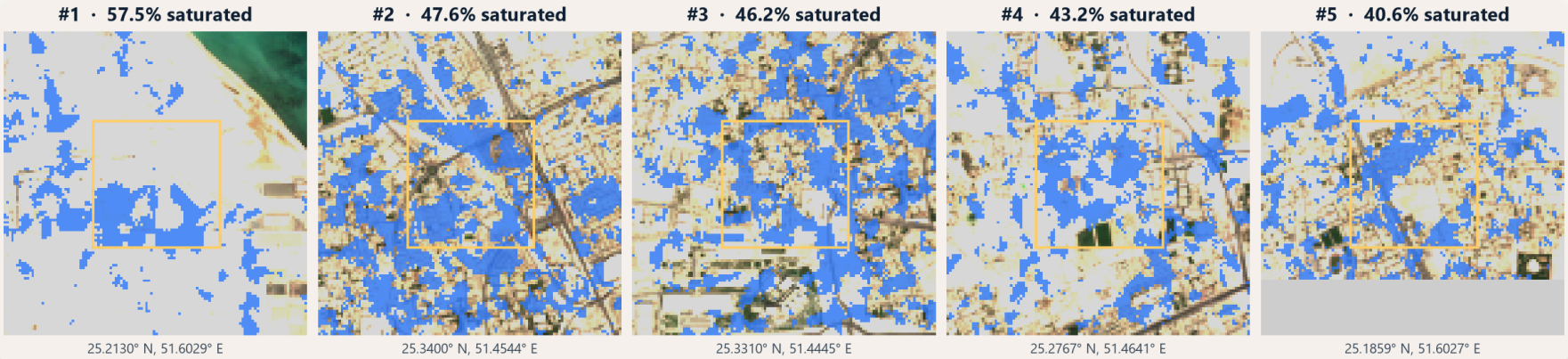
1 to 5: top priority cells for inspection

Live interactive map
an0m1z3r.github.io/doha-flood-watch

Live: an0m1z3r.github.io/doha-flood-watch/results/doha_flood_impact_map.html

PRIORITY LIST

The five 1 km cells to inspect first



Yellow box

The ranked 1 km cell

Blue

Ground saturated by the storm

Full list

Top 15 with coordinates in results/top15_priority_cells.csv

KEY NUMBERS

Main results

57.5 km²

rain-saturated ground, 15.9% of the land analysed

12.6×

signal vs false alarms on a dry week (57.5 vs 4.5 km²)

42.2 km²

of built-up Doha saturated, 17.9% of all built-up land

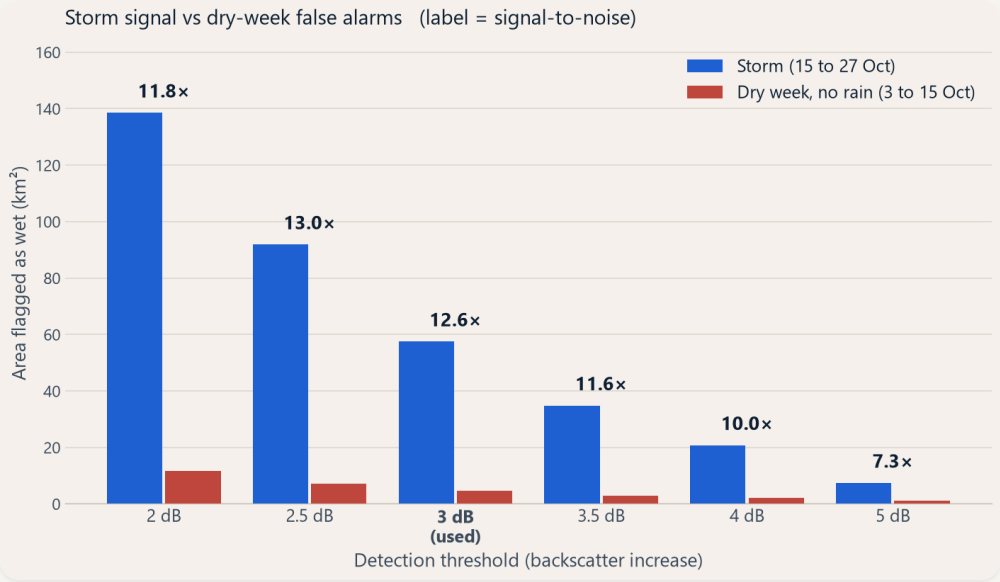
Top 15

1 km cells ranked for inspection; the worst had 57.5% of its land saturated

Area analysed: 361 km² of land in Doha, Al Rayyan and northern Al Wakrah

VALIDATION AND LIMITATIONS

Checking the result



Limitations

- The first radar pass came 7 days after the storm, so we can only map saturated ground
- Standing water at day 7 (0.08 km²) is not above noise (0.06 km²), so we don't claim it
- Dense streets: building echoes can hide water
- Land cover from 2021, event in 2018
- Not yet checked against Ashghal hotspot records

No official flood map exists for this event, so the dry-week control is our ground reference

IMPACT AND BUSINESS MODEL

What it changes, and who pays

Impact

- Crews and pumps go to the worst areas first
- Repeat-flooding hotspots guide drainage investment
- SDG 11 (resilient cities) and SDG 13 (climate action)
- Supports Qatar National Vision 2030 environmental goals

Business model

- Annual subscription for government agencies: automatic report after every heavy-rain event
- Per-event reports and a flood-risk score per location for insurers
- Data costs almost nothing, since Copernicus data is free

TEAM

Our team

Ali AlHenzab

Team lead & business

Business model, stakeholder contact, pitch

Hamad Abusheraida

Satellite data analyst

Sentinel-1 radar processing, flood detection, validation

Abdulla AlMuhaza

Software & data engineer

Notebook, code, repository, interactive map

Fahad AlMuraikhi

Impact & user research

Qatar flood context, user interviews, SDG and Vision 2030 alignment

Between us we cover the satellite analysis, the software, the users and the business side.

NEXT STEPS

What we would do in incubation

Faster revisits

Sentinel-1A, 1C and commercial radar together: images within 1 to 3 days, early enough to see standing water

Hotspot atlas

Every major Qatar rain event since 2015: cells that flood every time

Hyperspectral

813 / EnMAP spectra to separate wet soil, water and dark urban surfaces

Deploy on GIQ

Triggered by Qatar Meteorology Department heavy-rain alerts; map delivered hours after the next pass

Thank you · Team Stark, Qatar

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